



Machine Learning

AIML CZG565

M8 : Bayesian Learning

BITS Pilani

Pilani Campus

Course Faculty of M.Tech Cluster
BITS – CSIS - WILP

Disclaimer and Acknowledgement



- These content of modules & context under topics are planned by the course owner Dr. Sugata, with grateful acknowledgement to many others who made their course materials freely available online
- We here by acknowledge all the contributors for their material and inputs.
- We have provided source information wherever necessary
- Students are requested to refer to the textbook w.r.t detailed content of the presentation deck shared over canvas
- We have reduced the slides from canvas and modified the content flow to suit the requirements of the course and for ease of class presentation

Slide Source / Preparation / Review:

From BITS Pilani WILP: Prof.Sugata, Prof.Chetana, , Prof.Monali, Prof.Rajavadhana

External: CS109 and CS229 Stanford lecture notes, Dr.Andrew NG and many others who made their course materials freely available online

Course Plan

M1	Introduction & Mathematical Preliminaries
M2	Machine Learning Workflow
M3	Linear Models for Regression
M4	Linear Models for Classification
M5	Decision Tree
M6	Instance Based Learning
M7	Support Vector Machine – Planned to be discussed in CS 15 or CS 16
M8	Bayesian Learning
M9	Ensemble Learning
M10	Unsupervised Learning
M11	Machine Learning Model Evaluation/Comparison

$MSE \rightarrow LReg.$
 $MLE \rightarrow Logit$

MLE

MAP

Bayesian Learning Parameter Estimation

- Where does the **cost** come from? - Logistic regression
- Why least-squares **cost** function, be a reasonable choice? – Linear regression

Distribution

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

Mileage (in kmpl)	Car Price (in cr)
Neutral	High
Less	Low
Neutral	Medium
More	Low

Mileage (in kmpl)	Car Price (in cr)
9.8	High
9.12	Low
9.5	High
10	Low

P.D.F

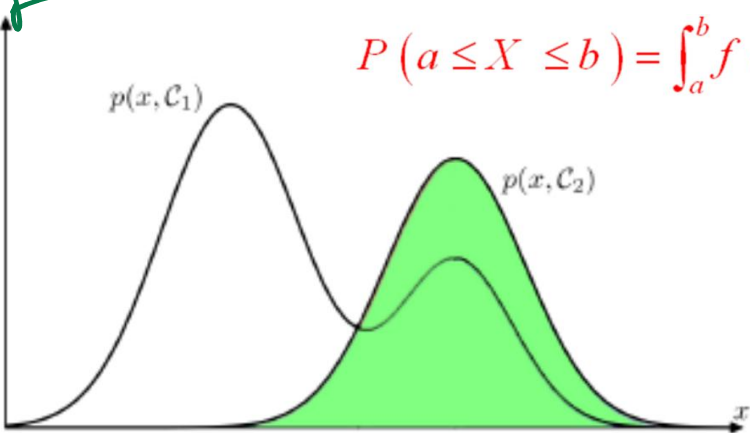
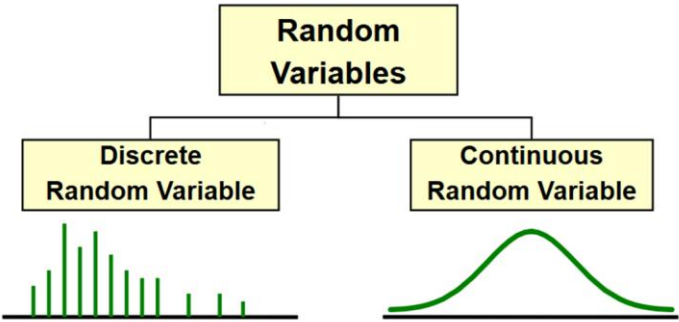
Represents a possible numerical value from a random event

$$p(X = x_i, Y = y_j) = \frac{n_{ij}}{N}$$

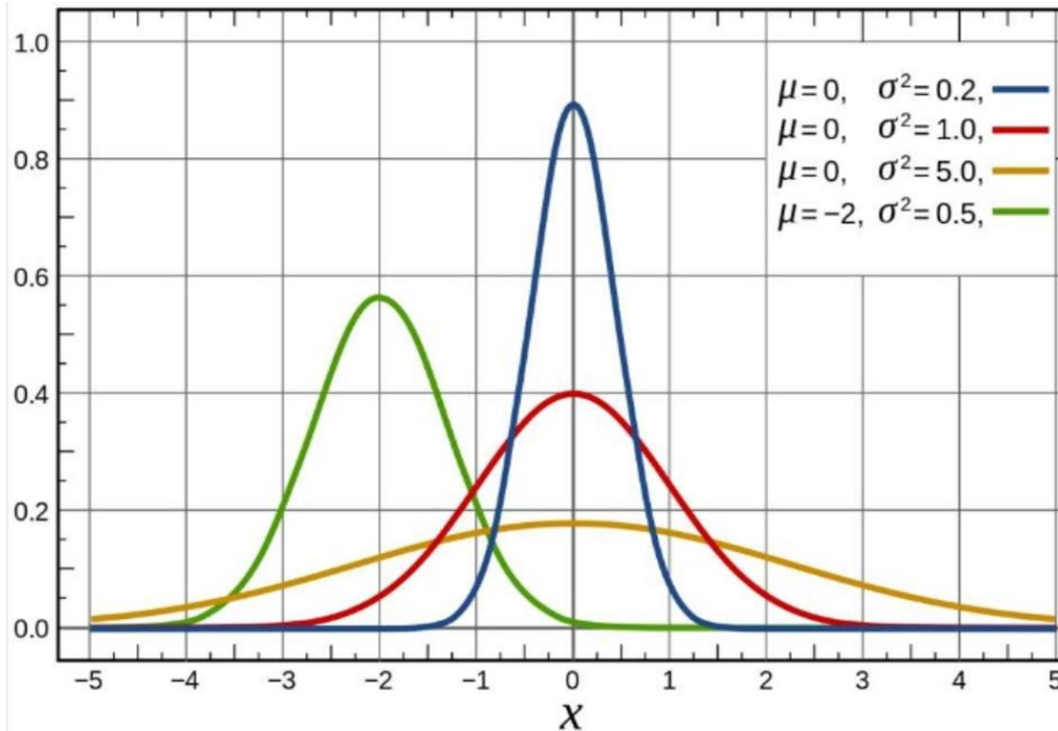
P.M.F

$$p(Y = y_j | X = x_i) = \frac{n_{ij}}{c_i}$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$



Parameter Estimation



Distribution	Parameters
Bernoulli(p)	$\theta = p$
Poisson(λ)	$\theta = \lambda$
Uniform(a,b)	$\theta = (a,b)$
Normal(μ, σ^2)	$\theta = (\mu, \sigma^2)$
$Y = mX + b$	$\theta = (m,b)$

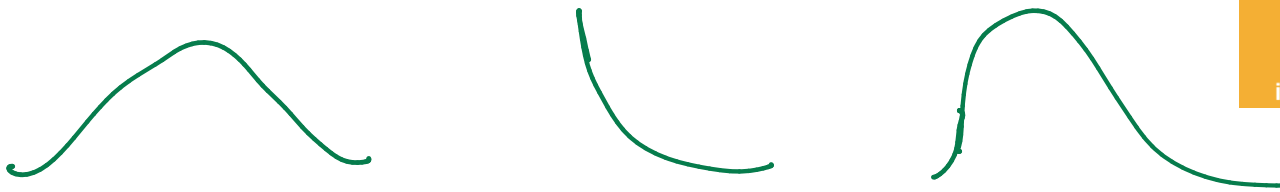
θ is the parameter of a distribution.

θ can be a vector of parameters

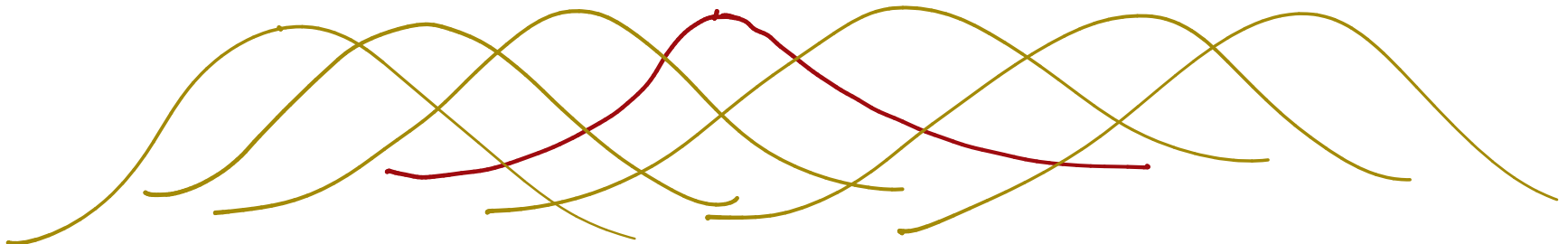
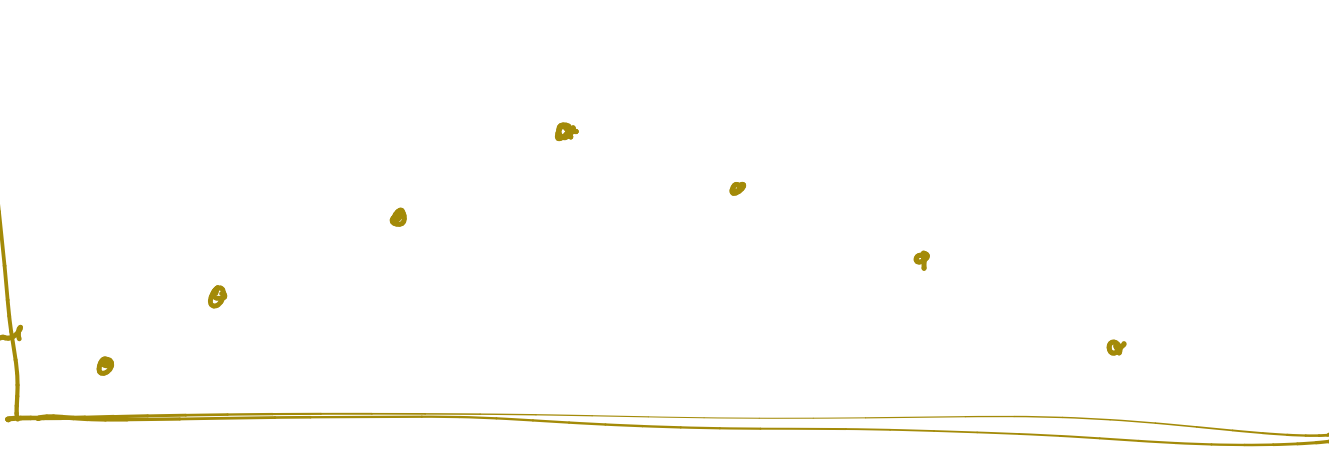
Distribution = model + parameter θ

Find $\theta = (\text{Mean}, \text{SD})$ from the data $\mathbf{X}_i$
 ie., $(x_i, P(x_i))$

$$N(x | \mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Likelihood
of
observing
data



Parameter in ML

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

$$\text{CarPrice} = 8.5 + \overset{\gamma_1}{0.5} \text{ Mileage} - \overset{\gamma_2}{1.5} \text{ Mileage}^2$$

Parameters : $(\theta_0, \theta_1, \theta_2)$

Mileage (in kmpl)	Car Price (in cr)
9.8	High
9.12	Low
9.5	High
10	Low

$$\text{CarPrice} = \frac{1}{1 + e^{-8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2}}$$

Parameter Estimation in ML

$$y = \theta^T x + \epsilon$$

innovate

achieve

lead

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

Assumption: Data are__

$$\text{CarPrice} = 8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2$$

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)}$$

$$\epsilon^{(i)} \sim N(0, \sigma^2)$$

$$P(D|\theta)$$

Parameters : $(\theta_0, \theta_1, \theta_2)$

Find $\theta = (\theta_0, \theta_1, \theta_2)$ from the data $\mathbf{X}_i$
ie., $(\text{Mileage}_i, \text{CarPrice}_i)$

- IID samples: $X_1 \dots X_n$ where all X_i are independent and have the same distribution.

- Either same PMF (discrete) or same PDF (continuous)

- $f(X|\theta)$

Likelihood of different values of X depends on the values of our parameters θ

$f(\cdot)$ is either PDF or PMF

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$$f(X_1, X_2, \dots, X_n | \theta)$$

hypo. are equally likely

Maximum A Posteriori (MAP)

choose the parameters θ that is the most likely, given the data

$$f(\theta | X_1, X_2, \dots, X_n)$$

Intuition of Bayes Theorem

MAP : $f(\theta | X_1, X_2, \dots, X_n)$

MLE: $f(X_1, X_2, \dots, X_n | \theta)$

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

$P(h)$ = prior probability of hypothesis h

$P(D)$ = prior probability of training data D

$P(h|D)$ = probability of h given D

$P(D|h)$ = probability of D given h

$L(\theta)$, probability of data given parameter θ

likelihood prior

posterior

$$P(\theta | \text{data}) = \frac{P(\text{data} | \theta) P(\theta)}{P(\text{data})}$$

After seeing data, posterior belief of θ

Before seeing data, prior belief of θ
e.g. what is distribution over parameters θ

Maximum Likelihood Estimation (MLE)

$$L(\theta)$$

1. Determine formula for $LL(\theta)$
2. Differentiate $LL(\theta)$ w.r.t. (each) θ
3. Solve

MLE – Linear Regression Model

Given the noise $\epsilon^{(i)}$ obeys a Normal distribution each $y^{(i)}$ must also obey a Normal distribution around the true target value

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51

$$\text{CarPrice} = 8.5 + 0.5 \text{ Mileage} - 1.5 \text{ Mileage}^2$$

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)}$$

$$\epsilon^{(i)} \sim N(0, \sigma^2)$$

Parameters : $(\theta_0, \theta_1, \theta_2)$

Find $\theta = (\theta_0, \theta_1, \theta_2)$ from the data $\mathbf{X}_i$

ie., (Mileage_i, CarPrice_i)

$$p(\epsilon^{(i)}) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(\epsilon^{(i)})^2}{2\sigma^2}\right)$$

$$y^{(i)} | x^{(i)}; \theta \sim N(\theta^T x^{(i)}, \sigma^2)$$

Maximum Likelihood Estimation (MLE)

select that parameters θ that make the observed data the most likely

$$f(X_1, X_2, \dots, X_n | \theta)$$

$$p(y^{(i)} | x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

MLE – Linear Regression Model

$$p(y^{(i)} | x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} \right)$$

1. Determine formula for $LL(\theta)$

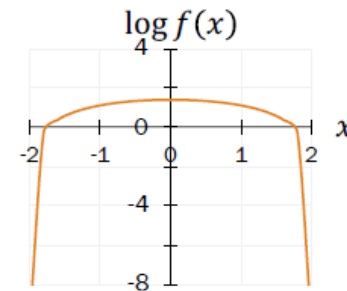
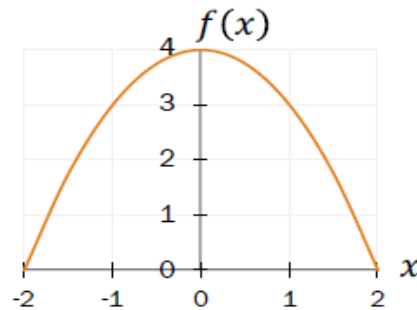
$$\begin{aligned} L(\theta) &= \prod_{i=1}^n p(y^{(i)} | x^{(i)}; \theta) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp \left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2} \right) \end{aligned}$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} L(\theta) \quad \rightarrow \quad LL(\theta) = \log L(\theta)$$

3. Solve

Mileage (in kmpl)	Car Price (in cr)
9.8	10.48
9.12	1.75
9.5	6.95
10	2.51



MLE answers the question: For which parameter value does the observed data have the largest probability?

MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n p(y^{(i)} | x^{(i)}; \theta) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \end{aligned}$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\Rightarrow \hat{\theta} = \underset{\theta}{\operatorname{argmax}} L(\theta) \quad \text{where} \quad LL(\theta) = \log L(\theta)$$

3. Solve

$$\begin{aligned} &\Rightarrow = \log L(\theta) \\ &= \log \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \\ &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \\ &= \left[n \log \frac{1}{\sqrt{2\pi}\sigma} \right] - \frac{1}{\sigma^2} \cdot \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2 \end{aligned}$$

Mileage (in kmpl)	Car Price (in cr)
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MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

$$\log L(\theta) = n \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2} \cdot \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$= -\frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

$$= -\frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

msl

3. Solve



MLE – Linear Regression Model

$$p(y^{(i)}|x^{(i)}; \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right)$$

1. Determine formula for $LL(\theta)$

$$\log L(\theta) = n \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2} \cdot \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ
- $$= -\frac{1}{2\sigma^2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$
- $$= -\frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2$$

3. Solve

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} L(\theta)$$
$$= \operatorname{argmax} \left(-\frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2 \right)$$
$$= \operatorname{argmin} \left(\frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T x^{(i)})^2 \right)$$

With probabilistic assumptions on the data, least-squares regression corresponds to finding the MLE of θ

MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^\top x_i))$$

1. Determine formula for $LL(\theta)$ $LL(\theta) = \log \prod_{i=1}^n P_\theta(y^{(i)} | x^{(i)}) = \sum_{i=1}^n \log P_\theta(y^{(i)} | x^{(i)})$
2. Differentiate $LL(\theta)$ w.r.t. (each) θ $p(y \mid x; \theta) = (h_\theta(x))^y (1 - h_\theta(x))^{1-y}$

3. Solve

Mileage (in kmpl)	Car Price (in cr)
9.8	High
9.12	Low
9.5	High
10	Low

Bernoulli MLE Estimation

$X_1, X_2, \dots, X_n$ where $X_i \sim \text{Ber}(p)$. PMF of a Bernoulli $p^{X_i}(1-p)^{1-X_i}$

$$P_\theta(Y = 1 | X = x) = h_\theta(x) = \frac{1}{1 + e^{-\theta^\top x}}$$

$$P_\theta(Y = 0 | X = x) = 1 - h_\theta(x) = \frac{e^{-\theta^\top x}}{1 + e^{-\theta^\top x}}$$

MLE answers the question: For which parameter value does the observed data have the largest probability?

MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^T x_i))$$

1. Determine formula for $LL(\theta)$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n P(Y = y^{(i)} \mid X = \mathbf{x}^{(i)}) \\ &= \prod_{i=1}^n \sigma(\theta^T \mathbf{x}^{(i)})^{y^{(i)}} \cdot [1 - \sigma(\theta^T \mathbf{x}^{(i)})]^{(1-y^{(i)})} \end{aligned}$$

CEL

$$LL(\theta) = \sum_{i=1}^n y^{(i)} \log \sigma(\theta^T \mathbf{x}^{(i)}) + (1 - y^{(i)}) \log[1 - \sigma(\theta^T \mathbf{x}^{(i)})]$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$\begin{aligned} \frac{\partial LL(\theta)}{\partial \theta_j} &= \frac{\partial}{\partial \theta_j} y \log \sigma(\theta^T \mathbf{x}) + \frac{\partial}{\partial \theta_j} (1 - y) \log[1 - \sigma(\theta^T \mathbf{x})] \\ &= \left[\frac{y}{\sigma(\theta^T \mathbf{x})} - \frac{1 - y}{1 - \sigma(\theta^T \mathbf{x})} \right] \frac{\partial}{\partial \theta_j} \sigma(\theta^T \mathbf{x}) \\ &= \left[\frac{y}{\sigma(\theta^T \mathbf{x})} - \frac{1 - y}{1 - \sigma(\theta^T \mathbf{x})} \right] \sigma(\theta^T \mathbf{x}) [1 - \sigma(\theta^T \mathbf{x})] \mathbf{x}_j \\ &= \left[\frac{y - \sigma(\theta^T \mathbf{x})}{\sigma(\theta^T \mathbf{x}) [1 - \sigma(\theta^T \mathbf{x})]} \right] \sigma(\theta^T \mathbf{x}) [1 - \sigma(\theta^T \mathbf{x})] \mathbf{x}_j \\ &= [y - \sigma(\theta^T \mathbf{x})] \mathbf{x}_j \end{aligned}$$
3. Solve

$$\theta := \theta + \alpha \nabla_{\theta} \ell(\theta)$$

MLE answers the question: For which parameter value does the observed data have the largest probability?

MLE – Logistic Regression

$$y_i \mid x_i \sim \text{Bern}(\sigma(\mathbf{w}^\top x_i))$$

1. Determine formula for $LL(\theta)$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n P(Y = y^{(i)} \mid X = \mathbf{x}^{(i)}) \\ &= \prod_{i=1}^n \sigma(\theta^T \mathbf{x}^{(i)})^{y^{(i)}} \cdot [1 - \sigma(\theta^T \mathbf{x}^{(i)})]^{(1-y^{(i)})} \end{aligned}$$

$$LL(\theta) = \sum_{i=1}^n y^{(i)} \log \sigma(\theta^T \mathbf{x}^{(i)}) + (1 - y^{(i)}) \log[1 - \sigma(\theta^T \mathbf{x}^{(i)})]$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ
- $$\frac{\partial LL(\theta)}{\partial \theta_j} = \sum_{i=1}^n [y^{(i)} - \sigma(\theta^T \mathbf{x}^{(i)})] x_j^{(i)}$$

3. Solve

MLE answers the question: For which parameter value does the observed data have the biggest probability?

MLE-Summary

- Consider a sample of n i.i.d. random variables $X_1, X_2, \dots, X_n$, drawn from a distribution $f(X_i|\theta)$
- θ_{MLE} maximizes the likelihood of data, $L(\theta)$ and $LL(\theta)$

$$L(\theta) = \prod_{i=1}^n f(X_i|\theta) \quad \hat{\theta} = \operatorname{argmax}_{\theta} L(\theta)$$

$$LL(\theta) = \log L(\theta) = \log \prod_{i=1}^n f(X_i|\theta) = \sum_{i=1}^n \log f(X_i|\theta)$$

$$\theta_{MLE} = \operatorname{argmax}_{\theta} LL(\theta)$$

① $L(\theta), LL(\theta)$
 ② der. $LL(\theta)$
 ③ solve

Additional Problems in MLE

MLE – Discrete - PMF

Example 1: Suppose that X is a discrete random variable with the following probability mass function: where $0 \leq \theta \leq 1$ is a parameter. The following 10 independent observations

X	0	1	2	3
$P(X)$	$2\theta/3$	$\theta/3$	$2(1-\theta)/3$	$(1-\theta)/3$

were taken from such a distribution: $(3,0,2,1,3,2,1,0,2,1)$. What is the maximum likelihood estimate of θ .

1. Determine formula for $LL(\theta)$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

3. Solve

$$LL(\theta) = 5 \log(1-\theta) + 5 \log \theta + c$$

$$\frac{\partial}{\partial \theta} LL(\theta) = 0 \Rightarrow \theta = ?$$

MLE – Discrete - PMF

1. Determine formula for $LL(\theta)$

$$L(\theta) = P(X=3)*P(X=0)*P(X=2)*.....*P(X=1)$$

$$= ((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

3. Solve

X	P(X)
3	$(1-\theta) / 3$
0	$2\theta / 3$
2	$2(1-\theta) / 3$
1	$\theta / 3$
3	$(1-\theta) / 3$
2	$2(1-\theta) / 3$
1	$\theta / 3$
0	$2\theta / 3$
2	$2(1-\theta) / 3$
1	$\theta / 3$

MLE – Discrete - PMF

1. Determine formula for $LL(\theta)$

$$L(\theta) = P(X=3) * P(X=0) * P(X=2) * \dots * P(X=1)$$

$$= ((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$LL(\theta) = \log [((1-\theta) / 3)^2 * (2\theta / 3)^2 * (2(1-\theta) / 3)^3 * (\theta / 3)^3]$$

$$= \log ((1-\theta) / 3)^2 + \log (2\theta / 3)^2 + \log (2(1-\theta) / 3)^3 + \log (\theta / 3)^3$$

$$= 2\log ((1-\theta) / 3) + 2\log (2\theta / 3) + 3\log (2(1-\theta) / 3) + 3\log (\theta / 3)$$

$$= 2(\log ((1-\theta)) - \log 3) + 2(\log (2\theta) - \log 3) + 3(\log (2(1-\theta)) - \log 3) + 3(\log (\theta) - \log 3)$$

$$\text{Gradient} (LL(\theta)) = -\frac{2}{(1-\theta)} + \frac{2}{\theta} - \frac{3}{(1-\theta)} + \frac{3}{\theta}$$

3. Solve

$$= \frac{-5\theta + 5 - 5\theta}{(1-\theta)\theta} = \frac{-10\theta + 5}{(1-\theta)\theta} = 0 \rightarrow \theta = 0.5$$

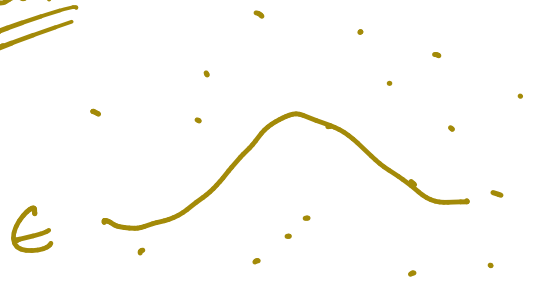
Additional Exercise for Student's Practice

Consider inputs x_i which are real valued attributes and the outputs y_i which are real valued of the form $y_i = f(x_i) + e_i$, where $f(x_i)$ is the true function and e_i is a random variable representing laplacian noise with PDF given by

$$f(y_i/\theta) = \frac{1}{2\theta} * e^{\frac{-|y_i - \mu|}{\theta}}$$

Implementing a linear regression model of the form, $h(x_i) = \sum_{i=0}^n \theta_i x_i$ and $\mu = h(x_i)$ find the maximum likelihood estimator of θ . Comment on the loss function.

low + ϵ plot
re. plot



ber

111



6-8pm



Additional Exercise for Student's Practice

1. Determine formula for $LL(\theta)$

$$L(\theta) = L(\theta) = \frac{1}{2\theta} e^{\frac{-|y_1 - \mu|}{\theta}} * \frac{1}{2\theta} e^{\frac{-|y_2 - \mu|}{\theta}} * \frac{1}{2\theta} e^{\frac{-|y_3 - \mu|}{\theta}} * \dots * \frac{1}{2\theta} e^{\frac{-|y_n - \mu|}{\theta}}$$

$$= \left(\prod_{i=1}^n \frac{1}{2\theta} e^{\frac{-|y_i - \mu|}{\theta}} \right)$$

2. Differentiate $LL(\theta)$ w.r.t. (each) θ

$$LL(\theta) = \ln \left(\prod_{i=1}^n \frac{1}{2\theta} e^{\frac{-|y_i - \mu|}{\theta}} \right) = -\ln(2\theta) \sum_{i=1}^n 1 + \sum_{i=1}^n \ln \left(e^{\frac{-|y_i - \mu|}{\theta}} \right)$$

$$= \underset{\theta}{\operatorname{argmax}} -n \ln(2\theta) - \sum_{i=1}^n \frac{|y_i - \mu|}{\theta}$$

$$= \underset{\theta}{\operatorname{argmin}} n \ln(2\theta) + \sum_{i=1}^n \frac{|y_i - \mu|}{\theta}$$

3. Solve

$$\text{Gradient} (LL(\theta)) = \frac{n}{(\theta)} - \frac{\sum_{i=1}^n |y_i - \mu|}{(\theta^2)} = 0$$

$$\frac{n}{(\theta)} = \frac{\sum_{i=1}^n |y_i - \mu|}{(\theta^2)}$$

$$\theta = \frac{\sum_{i=1}^n |y_i - \mu|}{n}$$

Instead of MSE, MAE is the maximum likelihood hypothesis. So MAE is appropriate for the loss function

References

- <https://www.inf.ed.ac.uk/teaching/courses/inf2b/learnnotes/inf2b-learn07-notes-nup.pdf>
- <https://cs229.stanford.edu/summer2019/cs229-notes2.pdf>
- Tom Mitchell – Chapter 6
- <https://nlp.stanford.edu/IR-book/pdf/13bayes.pdf>

Thank you !

Required Reading for completed session :

T1 - Chapter # 8 (Tom M. Mitchell, Machine Learning)

T1 - Chapter # 6 (Tom M. Mitchell, Machine Learning)

R1 – Chapter # 3 (Christopher M. Bhisop, Pattern Recognition & Machine Learning)

& Refresh your MFDS & ISM parallel course basics

Next Session Plan :

Module 9 : Ensemble Learning